

WRENTHAM AGDM, AB, Canada

WMO: 713430

Lat: 49.4952N

Lon: 112.1116W

Elev: 944

StdP: 90.49

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-27.0	-23.7	-30.3	0.3	-26.8	-26.8	0.4	-23.5	16.3	2.0	15.1	2.7	3.2	290	0.725

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.1	31.3	16.0	29.5	15.7	27.6	15.3	18.5	26.0	17.5	25.6	16.4	25.0	4.4	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.9	12.7	22.1	14.5	11.5	20.6	13.2	10.6	19.1	56.4	25.8	52.8	25.5	49.4	25.0	24.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.4	11.0	-31.4	34.8	3.6	1.9	-34.0	36.2	-36.1	37.3	-38.1	38.4	-40.7	39.8		
(5)			-31.6	21.2	3.6	1.7	-34.1	22.4	-36.2	23.4	-38.2	24.3	-40.8	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	-5.0	-5.8	0.3	5.7	10.9	14.9	18.3	17.8	12.9	6.3	-0.6	-6.0	(6)
(7)		DBStd	10.61	8.02	8.22	7.05	4.95	4.60	3.26	2.86	3.54	4.86	5.66	7.38	8.14	(7)
(8)		HDD10.0	2376	464	443	302	141	45	2	0	1	24	138	321	496	(8)
(9)		HDD18.3	4636	722	676	558	378	233	112	36	53	172	373	568	754	(9)
(10)		CDD10.0	874	0	0	2	13	73	148	258	243	111	23	3	0	(10)
(11)		CDD18.3	92	0	0	0	0	2	8	36	37	9	0	0	0	(11)
(12)		CDH23.3	1779	0	0	1	8	84	155	629	678	216	9	0	0	(12)
(13)		CDH26.7	579	0	0	0	1	15	28	210	255	69	1	0	0	(13)
(14)	Wind	WSAvg	5.1	6.0	5.1	5.6	5.5	5.3	4.9	3.9	3.9	4.7	5.4	5.7	5.6	(14)
(15)	Precipitation	PrecAvg	356	11	10	17	33	49	83	30	33	35	19	13	12	(15)
(16)		PrecMax	524	19	21	39	60	102	251	66	77	84	57	32	22	(16)
(17)		PrecMin	204	6	4	6	13	14	11	1	4	6	4	4	5	(17)
(18)		PrecStd	81	3	5	8	16	25	57	20	21	22	13	7	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.6	13.2	18.0	24.3	28.5	29.7	33.5	33.9	31.6	24.7	18.2	11.2	(19)
(20)			MCWB	5.9	5.6	8.0	11.0	13.7	17.4	16.8	16.0	14.7	12.1	8.8	4.7	(20)
(21)		2%	DB	8.5	9.1	15.1	20.8	25.9	27.0	31.0	31.7	28.8	20.8	13.6	8.0	(21)
(22)			MCWB	4.1	3.6	6.9	9.5	12.7	15.6	16.9	15.6	13.7	10.6	6.8	3.4	(22)
(23)		5%	DB	6.4	6.6	12.5	17.6	23.0	25.0	29.1	29.6	25.8	17.6	11.0	6.1	(23)
(24)			MCWB	2.6	2.2	5.6	8.0	11.6	14.9	16.9	15.1	12.9	9.1	5.4	2.3	(24)
(25)		10%	DB	4.7	4.6	9.9	14.6	20.3	22.9	27.0	27.5	22.6	14.7	8.6	4.2	(25)
(26)			MCWB	1.8	1.2	4.3	6.5	10.4	13.9	16.7	14.8	11.9	7.7	4.0	1.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	6.0	8.8	11.6	14.6	19.4	21.5	18.7	16.0	12.9	9.6	5.8	(27)
(28)			MCDWB	11.0	11.6	16.3	22.1	26.4	26.1	28.0	26.4	25.6	22.9	16.1	9.8	(28)
(29)		2%	WB	4.3	3.9	7.4	9.9	13.1	17.3	19.2	17.2	14.6	11.1	7.3	3.7	(29)
(30)			MCDWB	8.2	8.5	14.2	19.5	23.3	24.8	26.4	27.0	26.0	19.2	13.1	7.6	(30)
(31)		5%	WB	2.9	2.5	5.9	8.5	12.1	15.7	18.1	16.3	13.5	9.5	5.7	2.4	(31)
(32)			MCDWB	6.1	6.3	11.9	16.6	21.5	23.0	25.8	26.4	24.5	16.7	10.6	5.7	(32)
(33)		10%	WB	1.8	1.3	4.4	7.0	11.1	14.5	17.2	15.5	12.3	8.2	4.2	1.2	(33)
(34)			MCDWB	4.6	4.4	9.5	13.7	18.8	21.5	25.1	25.3	21.9	14.0	8.3	4.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	10.0	10.8	11.6	12.7	12.7	12.4	15.1	15.6	13.9	12.3	10.8	9.8	(35)
(36)			MCDBR	10.3	12.5	15.1	18.0	17.3	16.1	19.0	20.0	19.3	16.0	13.5	10.8	(36)
(37)			MCWBR	7.1	8.1	8.5	8.7	7.1	7.4	8.2	7.3	7.7	7.8	8.0	7.4	(37)
(38)		5% WB	MCDWB	10.2	12.2	14.5	16.7	15.6	14.5	16.2	17.0	18.0	15.4	13.2	10.7	(38)
(39)			MCWBR	7.3	8.4	8.6	8.5	6.8	7.5	8.2	7.0	7.4	7.8	8.1	7.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.256	0.264	0.285	0.323	0.338	0.336	0.338	0.355	0.317	0.285	0.258	0.244	(40)	
(41)		taud	2.462	2.488	2.486	2.382	2.387	2.441	2.445	2.367	2.450	2.522	2.503	2.446	(41)	
(42)		Ebn at Noon	826	907	938	925	921	922	914	880	886	865	817	790	(42)	
(43)		Edh at Noon	59	73	88	109	114	109	107	110	91	70	56	52	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.26	2.26	3.53	4.88	5.80	6.46	7.08	5.65	4.01	2.51	1.41	1.00	(44)	
(45)		RadStd	0.11	0.17	0.28	0.33	0.47	0.44	0.37	0.35	0.33	0.25	0.10	0.09	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (60 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page